A Contour-Stable Criterion for Uniform Quotient Tails

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Abstract

We give a sufficient continuum theorem for the orthogonal quotient route. Hilbert–Schmidt convergence, a common isolating contour with uniform resolvent control, and a contractive power of the limiting quotient imply stable selected subspaces and uniform RH-246 block constants. The current archive supplies fixed-noise identities and finite contractions but none of the four continuum hypotheses. The result is conditional and does not prove nonuniformity of the actual family.

1 Riesz and orthogonal projection stability

Let $A_s \in \mathcal{S}_2(H)$ satisfy $A_s \to A_0$ in $\mathcal{S}_2$. Suppose one positively oriented rectifiable Jordan contour Γ lies in every relevant resolvent set and isolates the same continued spectral cluster of finite algebraic multiplicity, and

$$\sup_{s,z\in\Gamma} \|(z-A_s)^{-1}\| \le M.$$

The Riesz projections $P_s = (2\pi i)^{-1}\int_\Gamma (z-A_s)^{-1}\,dz$ obey, by the resolvent identity,

$$\|P_s - P_0\| \le \frac{|\Gamma|}{2\pi} M^2 \|A_s - A_0\|. \tag{1}$$

Put $r = \operatorname{rank} P_0 < \infty$. For small s, $\|P_s - P_0\| < 1$, so $\operatorname{rank} P_s = r$. Let Π_s be the orthogonal projection onto $\operatorname{Ran} P_s$ and put $R_s = I - \Pi_s$. Finite-rank gap stability gives $\Pi_s \to \Pi_0$. Choose the standard near-identity unitary $U_s \to I$ with $U_s R_0 U_s^* = R_s$. On the fixed space $R_0 H$ define

$$\widetilde{C}_s = U_s^* R_s A_s R_s U_s\big|_{R_0 H}.$$

This is unitarily equivalent to the orthogonal quotient compression on $R_s H$, so its power traces and Schatten norms are unchanged.

Theorem 1.1 (Contour-stable quotient criterion)**.** *Under the preceding hypotheses, $\widetilde{C}_s \to \widetilde{C}_0$ in $\mathcal{S}_2$. If for some $m \ge 2$,*

$$\|\widetilde{C}_0^m\| = \eta_0 < 1,$$

then on a sufficiently small parameter neighborhood there are uniform constants

$$K_m = \sup_s \|\widetilde{C}_s^m\|_1 < \infty, \qquad \eta_m = \sup_s \|\widetilde{C}_s^m\| < 1, \qquad L_r = \sup_s \|\widetilde{C}_s^r\| < \infty \quad (0 \le r < m).$$

Proof. Equation (1) and finite-rank gap stability give the projections and near-identity unitaries above. The ideal property yields

$$\|R_s A_s R_s - R_0 A_0 R_0\|_2 \le \|A_s - A_0\|_2 + \|R_s - R_0\|(\|A_s\|_2 + \|A_0\|_2),$$

and hence

$$\|\widetilde{C}_s - \widetilde{C}_0\|_2 \le \|R_s A_s R_s - R_0 A_0 R_0\|_2 + 2\|U_s - I\|\,\|R_s A_s R_s\|_2 \longrightarrow 0.$$

Set $B_2 = \sup_s \|\widetilde{C}_s\|_2$ and $B = \sup_s \|\widetilde{C}_s\|$ on a small neighborhood. Then

$$L_0 = 1, \qquad L_r \le B^r, \qquad K_m \le B_2^2 B^{m-2}.$$

The power telescoping identity also gives

$$\|\widetilde{C}_s^m - \widetilde{C}_0^m\| \le m B^{m-1} \|\widetilde{C}_s - \widetilde{C}_0\|.$$

After shrinking the neighborhood, $\eta_m \le (1+\eta_0)/2 < 1$, proving all three uniform bounds. $\square$

Corollary 1.2. *For every R with $\eta_m R^m < 1$, the RH-246 block-power estimate holds uniformly on that neighborhood:*

$$\sum_{n \ge m} \frac{|\operatorname{Tr}(\widetilde{C}_s^n)| R^n}{n} \le \frac{K_m R^m}{m(1 - \eta_m R^m)} \sum_{r=0}^{m-1} L_r R^r.$$

In particular the corresponding logarithmic quotient tail is uniform [2].

2 Archived hypothesis audit

RH-245 proves the fixed-noise orthogonal quotient identity [1]; RH-259 supplies 23 finite twelfth-power contractions [3]. The archived theorem flags nevertheless leave all four continuum inputs unavailable: no $\mathcal{S}_2$ convergence to a limiting family, common isolating contour, uniform resolvent bound, or contractive limiting quotient power is certified. Selected-subspace uniform stability and complete endpoint coverage are also false/open.

Thus the criterion is not activated. This is a scoped non-activation and missing-certificate result, not a proof that the underlying family violates the criterion. Even future activation would close only the local uniform quotient-tail obligation for the specified selector; it would not supply a legal anchored head or cloud-to-target coefficient bridge. The latter and Gates A–E remain false/open, with no Hilbert–Polya, zeta-divisor, or RH claim.

References

[1] Bin Wang. Orthogonal quotient superloop compression. *Preprint*, 2026. RH-245 preprint.

[2] Bin Wang. Block-power quotient envelope criterion. *Preprint*, 2026. RH-246 preprint.

[3] Bin Wang. Extended quotient block-power diagnostic. *Preprint*, 2026. RH-259 preprint.